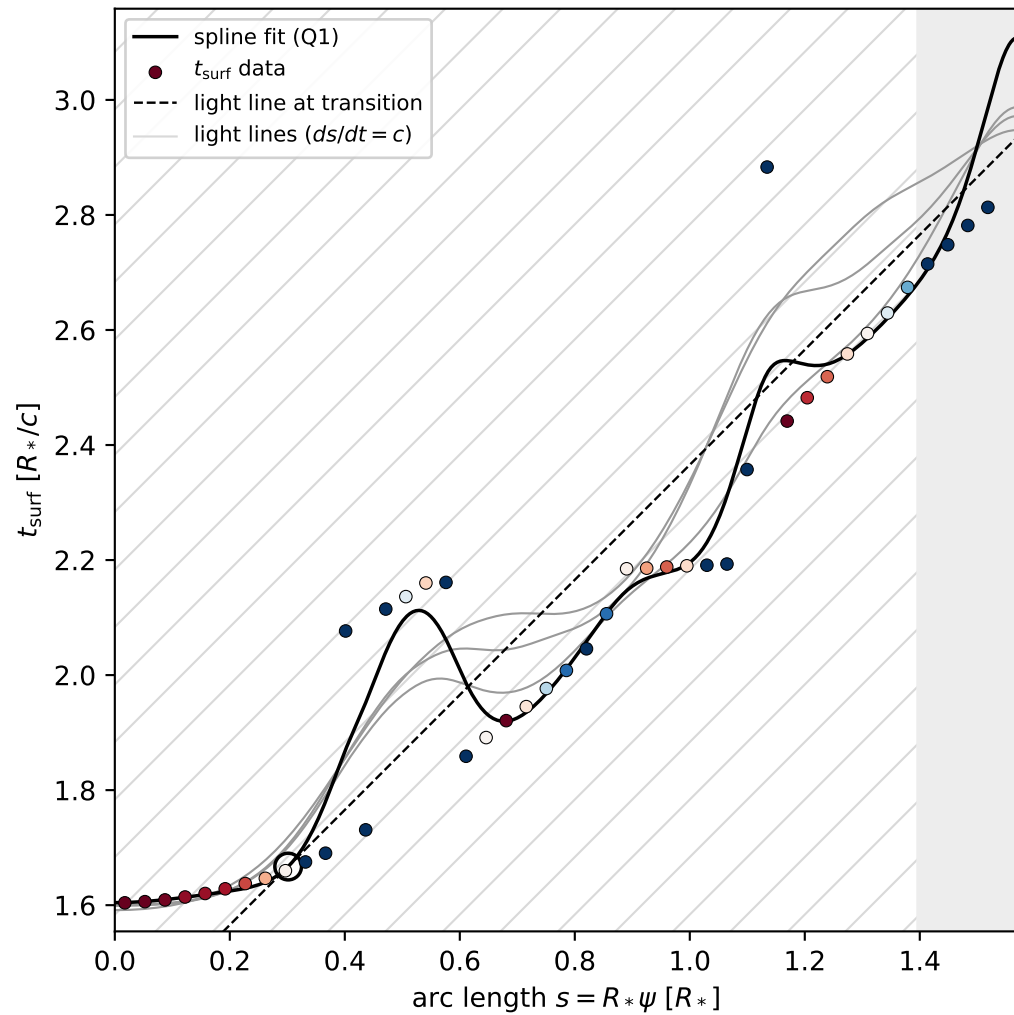


(a) surface arrival spacetime diagram



(b) pattern speed along the surface

